TRAISIM:
Real-Time Power System Simulation for Operator Training

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Cyprus University of Technology
Sustainable Power Systems Lab

Rte

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Presentation outline

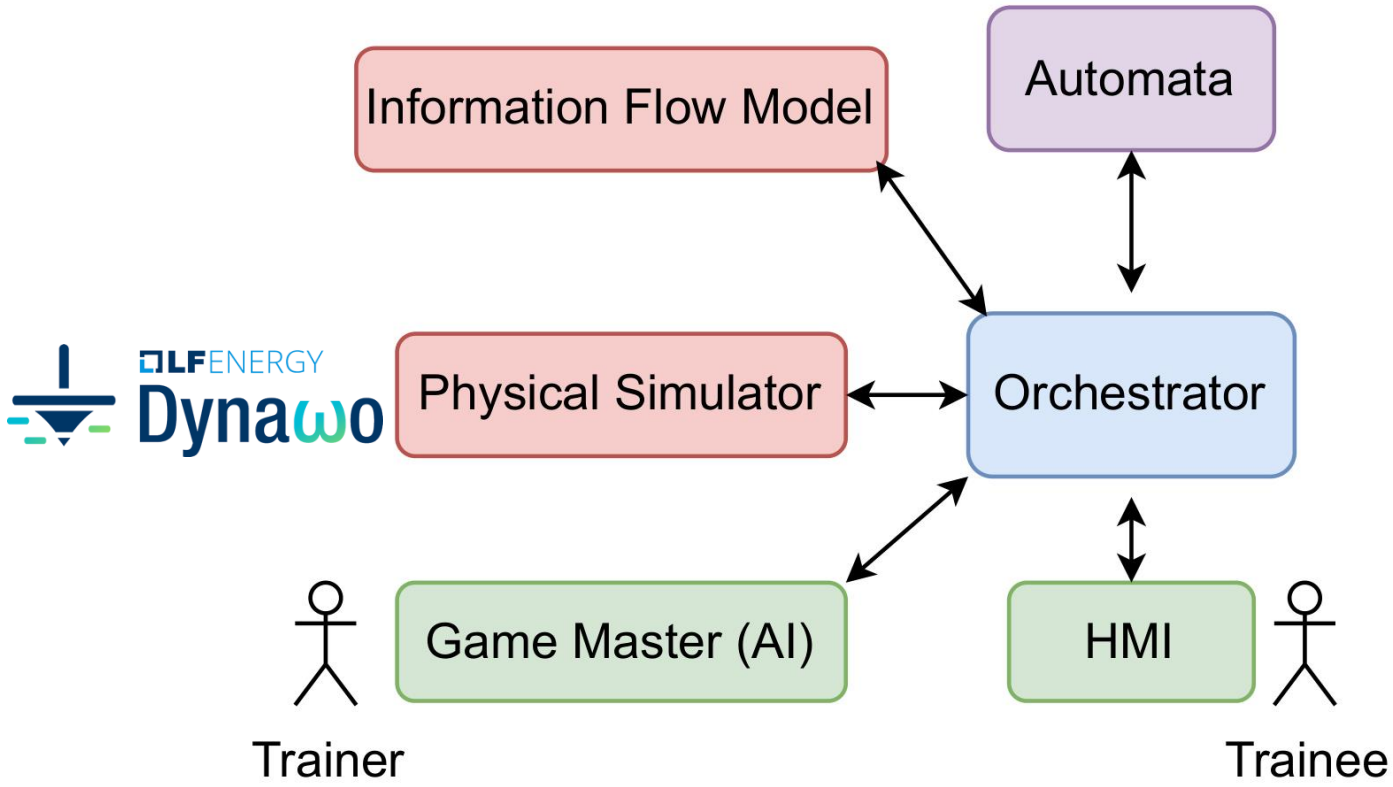
1. Real-time operator training: the system and its timing limits
2. Adaptive Model Selection and results
3. Numerical solver improvements and results
4. Conclusions and future work

1. **Real-time operator training: the system and its timing limits**
2. Adaptive Model Selection and results
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Training feedback must keep pace with the exercise

- Operators practise disturbances, switching and corrective actions.
- Simulator outputs must keep pace with SCADA updates, typically every 2-5 s.
- A 1 second solve budget leaves time for orchestration, communication and the operator interface.



Headroom = 1 s - step solve time

(Negative headroom indicates a missed deadline)

A large transmission-system model

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French transmission system, 63–400 kV

- More than 6,000 buses and 7,075 branches
- 4,009 substations and 3,039 dynamic loads
- About 370 detailed synchronous generators
- 839 tap changers across 2,138 transformers
- Multiple local and wide-area controllers

What the simulator has to solve

- Continuous dynamics of machines and controls
- Network equations balanced at every step
- Switching logic that changes with events

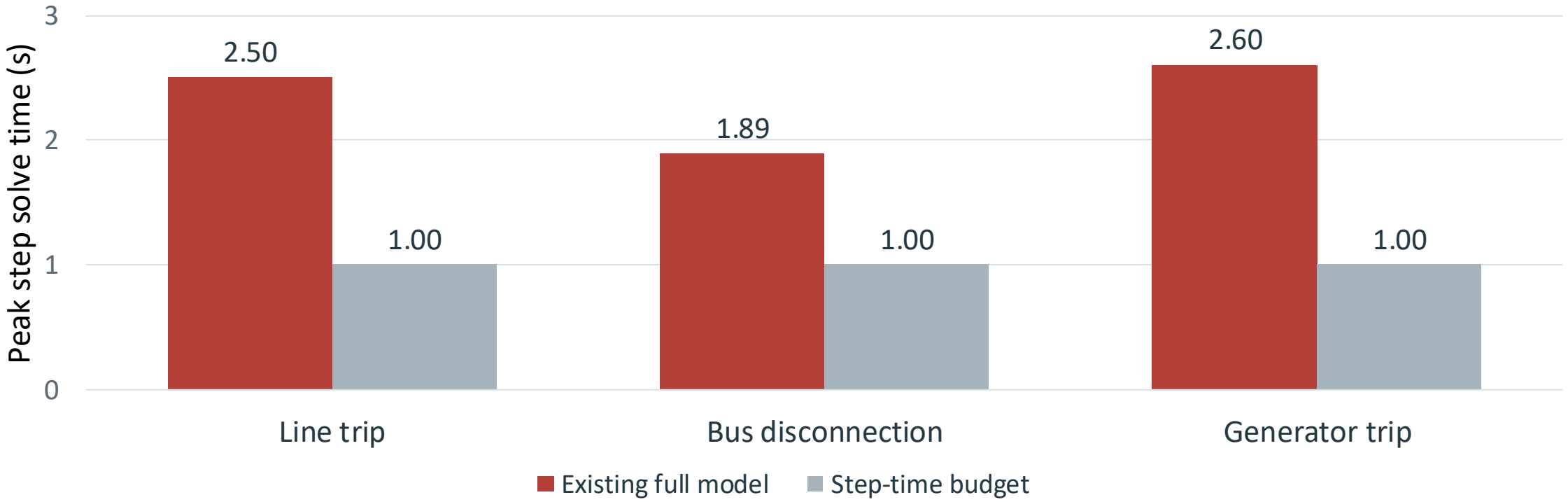
Together: a model whose equation set changes when the grid switches, dynamic model hits limits, or discrete controllers act.

374,989 continuous variables / 397,301 discrete variables / 353,160 event/root functions

(System size and generator counts vary with the initial operating point)

A disturbance can change both the equations and the switching state

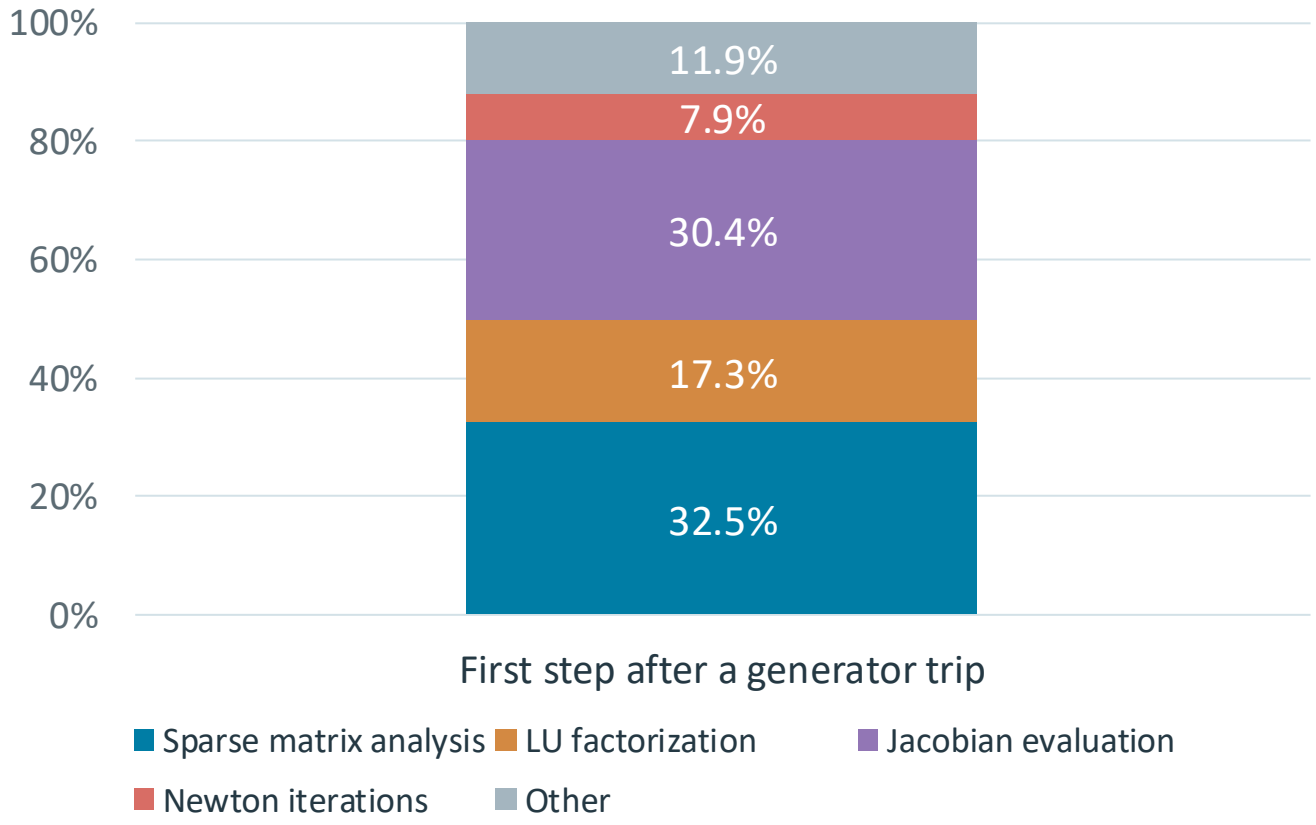
Existing Dynawo solver: missed real-time deadlines



Each late step is a pause on the trainee's screen, and the pauses cluster right after the disturbance, when the operator needs the response

Illustrative cases on the French transmission system.

Why disturbances are expensive to solve?



Profile of the first step after a generator trip in a detailed benchmark.

About 91% of solve time lies in the nonlinear equation solver

- Topology changes require a new consistent algebraic state.
- Sparse-matrix analysis and Jacobian evaluation dominate this profiled event step.

Two complementary routes to real time

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Two levers on the cost of a step: how many equations, and how much work each one takes

Part 2 of 4: a smaller model

Adaptive Model Selection

- **What changes**: detailed modelling is kept only where the scenario needs it, decided before the simulation.
- **Benefit**: fewer variables, equations and event functions in the model the solver sees.

Part 3 of 4: a faster solver

Numerical solver improvements

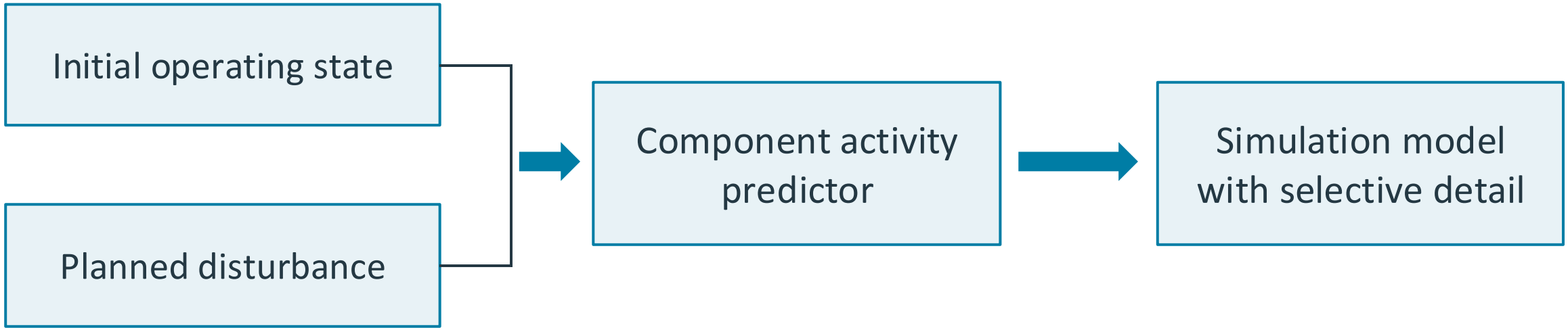
- **What changes**: how the solver treats the Jacobian, the threads and the restoration after an event.
- **Benefit**: less computation per step, with the model and its accuracy unchanged.

Timely feedback with the required dynamic fidelity:
each simulated second in less than a second of real time

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Adaptive Model Selection

Idea: Before simulation, predict where detailed modeling is needed and reduce the system model

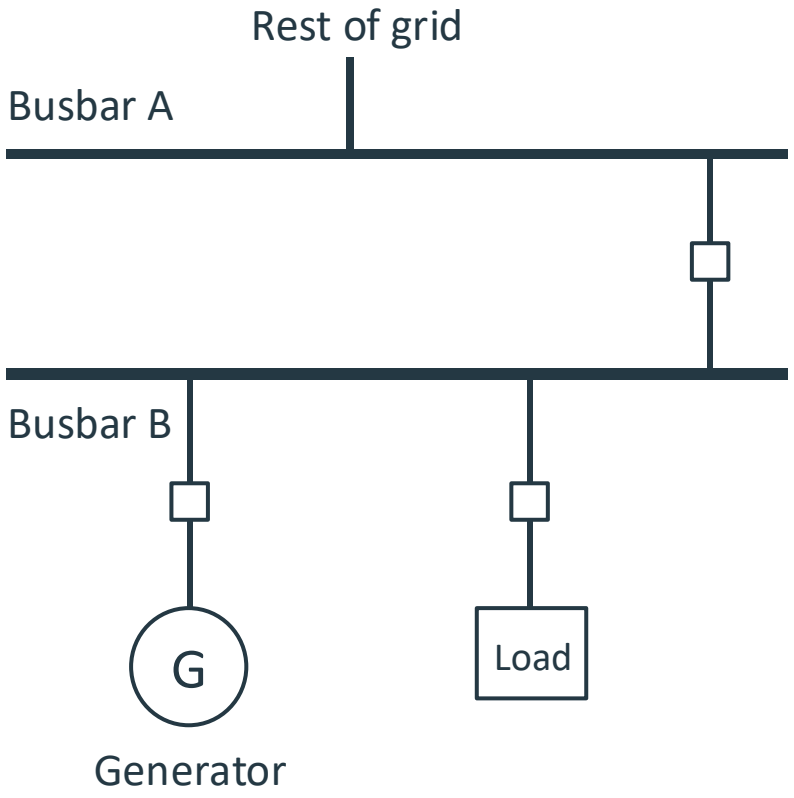


- Retain full detail at substations predicted to need it (high dynamic activity) and reduce the detail in other substations (low dynamic activity).
- Construct and solve a physical dynamic model for the chosen scenario.

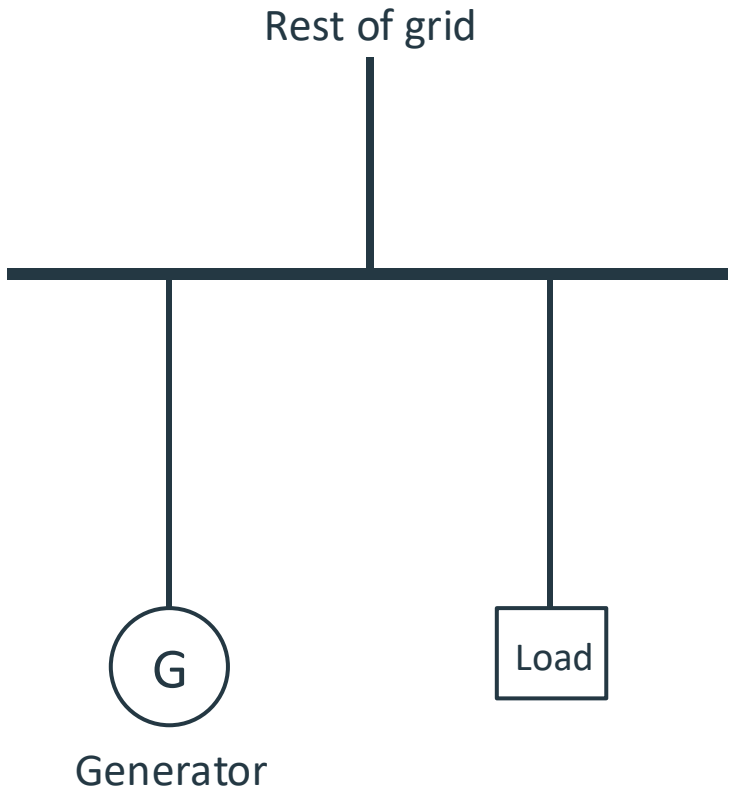
What can be simplified?

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Detailed switching topology



Equivalent bus connectivity



Same electrical connectivity while switch states remain fixed

Illustrative closed-switch substation. Full detail remains wherever switching or control actions matter.

From the planned event to a lighter model

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One pass before the scenario starts; the scenario itself runs as before

Starting point and planned event

the operating point of the day and the disturbance the trainer has scheduled

Activity predictor

a graph model, trained once per network on detailed offline simulations; about 1.4 ms per scenario

Keep detail where it is needed

full switching detail at substations predicted active or acting; simpler equivalents elsewhere

Simulate the lighter model

the same simulator, with 72 % fewer continuous and 66 % fewer discrete variables

- Uses only what is known before the disturbance.
- The retention threshold sets the balance between speed and accuracy.
- The whole pass is released as open-source software: github.com/SPS-L/DYNAGNN

Case studies and evaluation

An operating point specifies the initial loading, generation and equipment status.

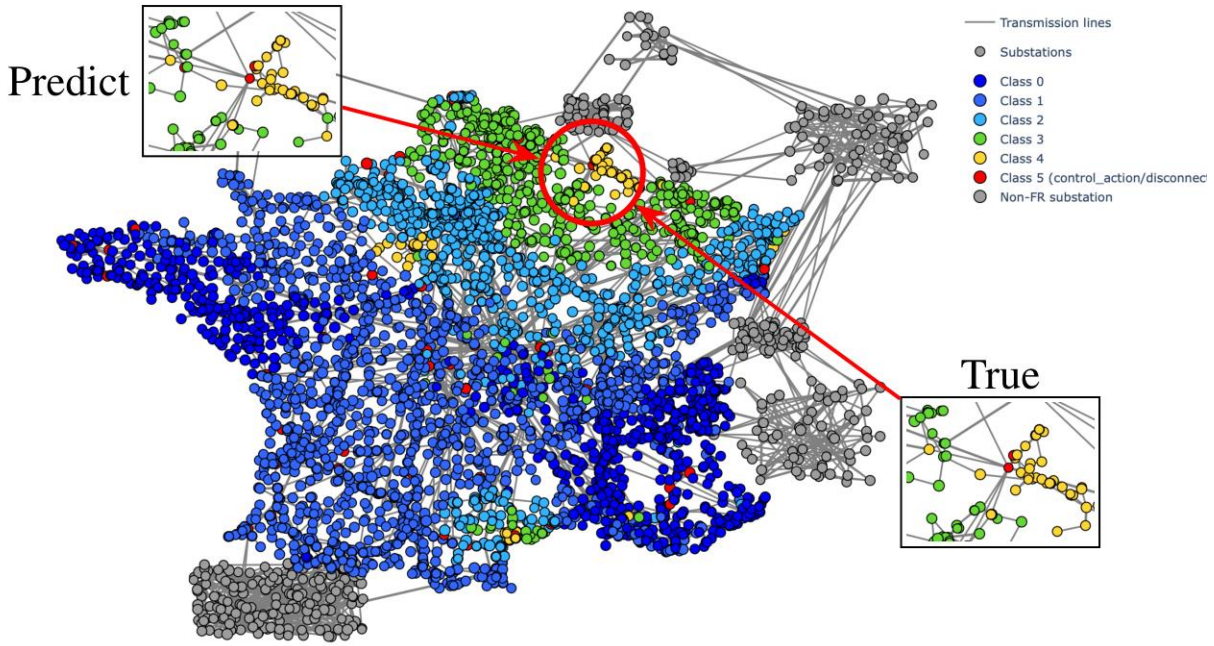
Quantity	Nordic 32	French transmission system
Buses	74	More than 6,000
Operating points	35	61
Usable scenarios	7,419	69,616
Training / validation / test	70 / 15 / 15 %	80 / 10 / 10 %
Evaluation role	Architecture comparison	Prediction at scale and AMS

- Feature scaling and activity thresholds use training data only.
- Test operating points are unseen. Component and event identities can recur.

Activity prediction on the French transmission system

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Held-out metric	Voltage	Apparent power
Exact accuracy	67.9%	79.2%
Within one class	93.0%	91.7%



Representative predicted versus true activity near an event. Metrics refer to the French held-out test set. Identifying the action flag that forces full detail is the weak spot, which the generator case shows.

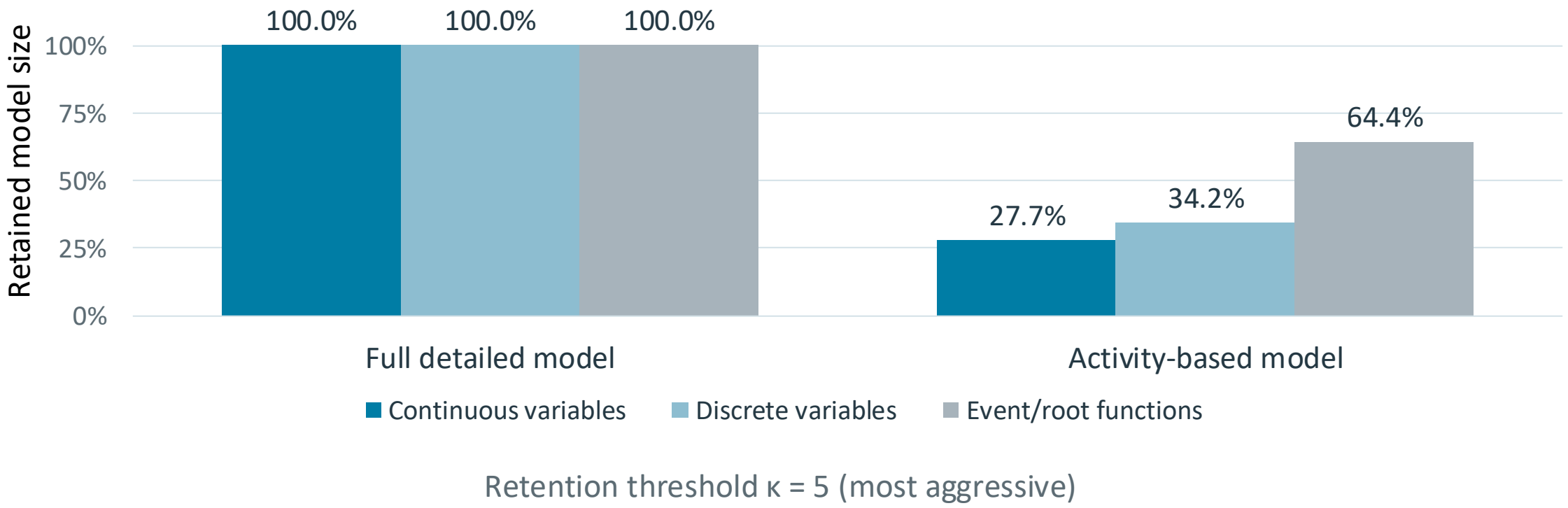
A small class error can still change the retain-or-simplify decision.

Fewer variables after selective simplification

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Keep full detail if any bus or generator class is at least κ

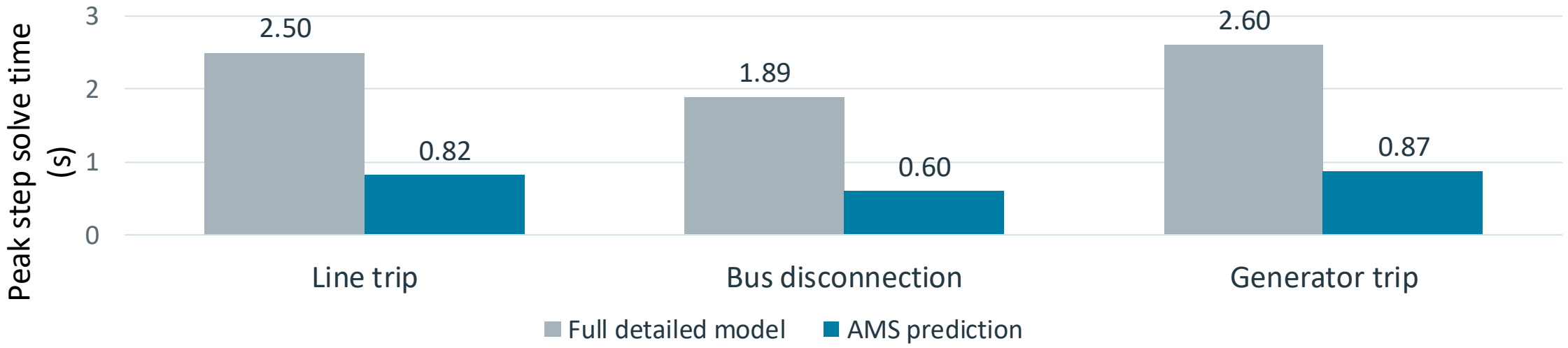
Elsewhere, replace detailed switch topology with equivalent bus connectivity.



Retention threshold κ = 5 (most aggressive)

Faster simulation, with fidelity checks

Faster steps must also preserve the voltage response



Case at κ = 5 (most aggressive)	AMS peak (s)	Late steps	Voltage error (pu)
Line: correct retention	0.8179	0	2.89×10^{-5}
Bus: extra detail retained	0.5978	0	1.07×10^{-5}
Generator: critical detail omitted	0.8672	0	**0.014**

Voltage error: maximum DTW RMSE across buses. Peak-step speedup ~3×

What this means for a training centre

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A lighter model built before the scenario, with a visible guard rail

What you get

- A lighter model, built before the scenario in milliseconds.
- Each simulated second in less than a second of real time in the illustrated cases.
- Same simulator, same scenario files.

What to watch

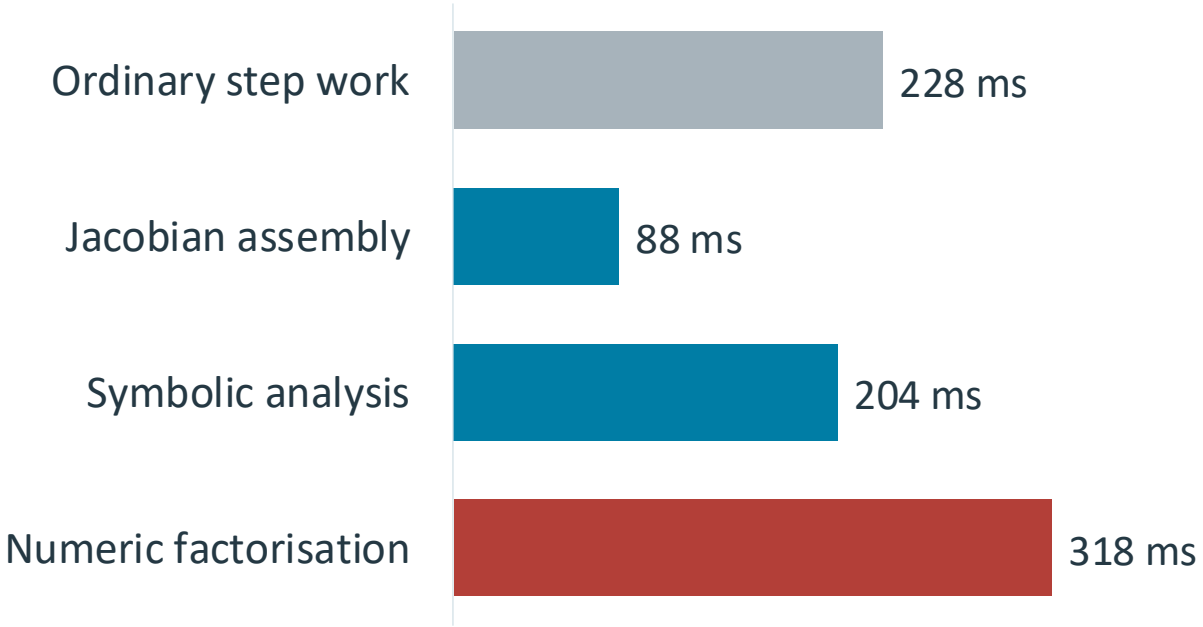
- A simplified critical substation shows as a voltage error, 0.014 pu.
- The retention threshold trades speed for safety.
- A new network needs its own training data.

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Where a late step spends its time

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Every late step contains a Jacobian evaluation



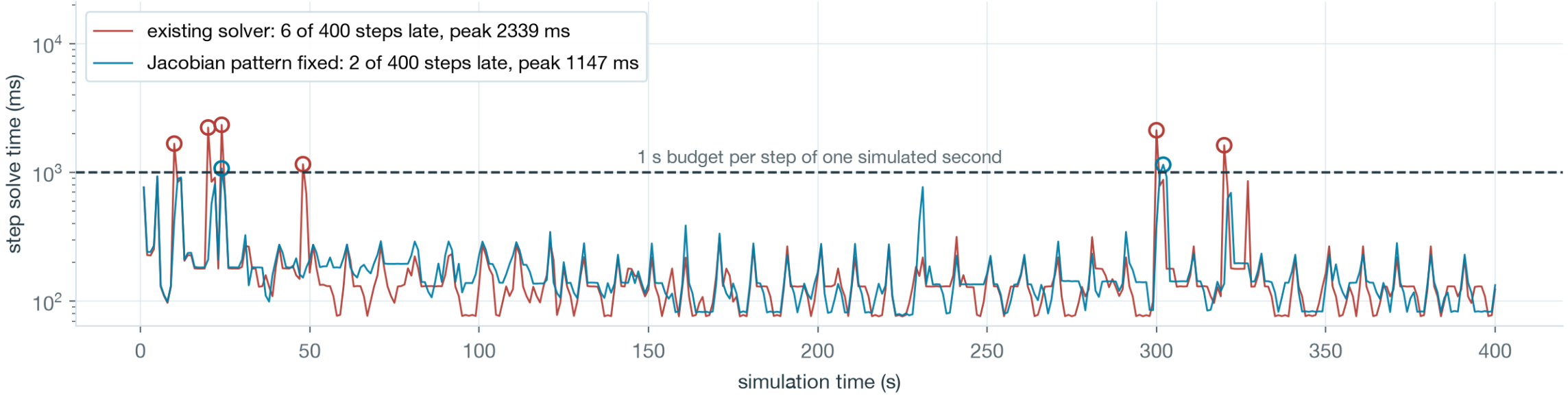
- A Jacobian evaluation costs about 624 ms, on top of about 228 ms of ordinary step work.
- A switching event forces a full rebuild, repeating the sparse-matrix analysis as well as the factorisation.
- Every one of the 613 late steps across 57 operating points of the French transmission system evaluates a Jacobian.

Three tested improvements target this step.

Keeping the Jacobian pattern fixed across switching

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A fixed sparsity pattern lets a switching event repeat only the numeric factorisation



	Existing	Pattern fixed	Reduction
400 s benchmark: late steps	6	2	67 %
400 s benchmark: peak step solve time	2339 ms	1147 ms	51 %
57 operating points: late steps	651	332	49 %

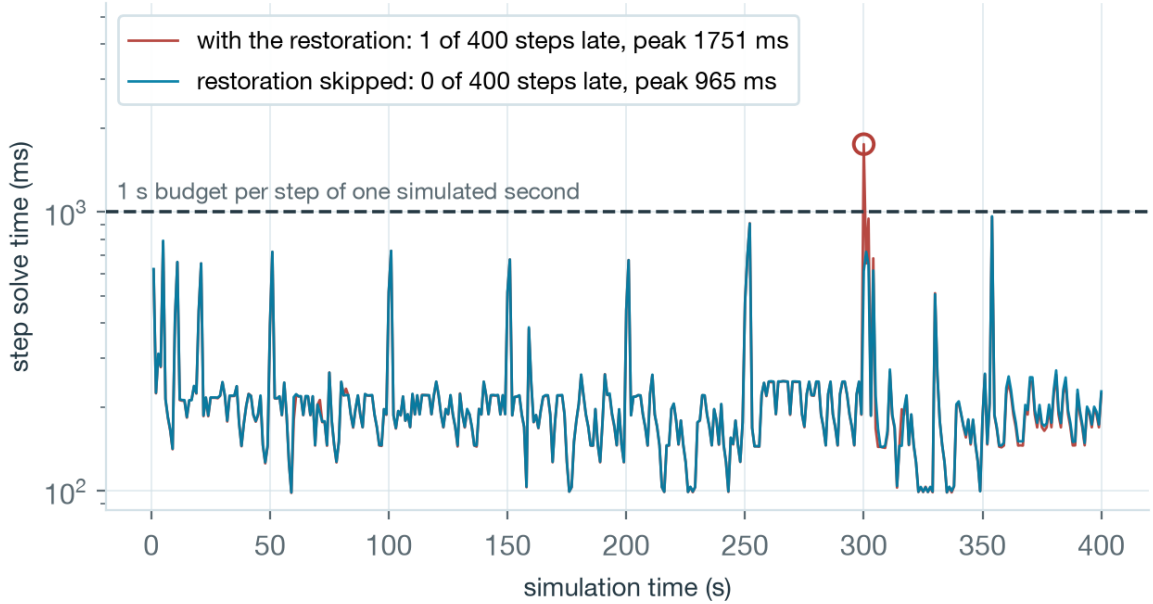
Affects switches, static lines and two-winding transformers that emit the same index set in every switch state.

Skipping the algebraic restoration after an event

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The fixed-step solver uses the restored values only as Newton's starting point

- After an event, the solver first re-solves the algebraic equations at the event time, a separate nonlinear solve.
- The next backward-Euler step does not use that result.
- Skipping it saves a nonlinear solve per event; events may then be detected a step apart.



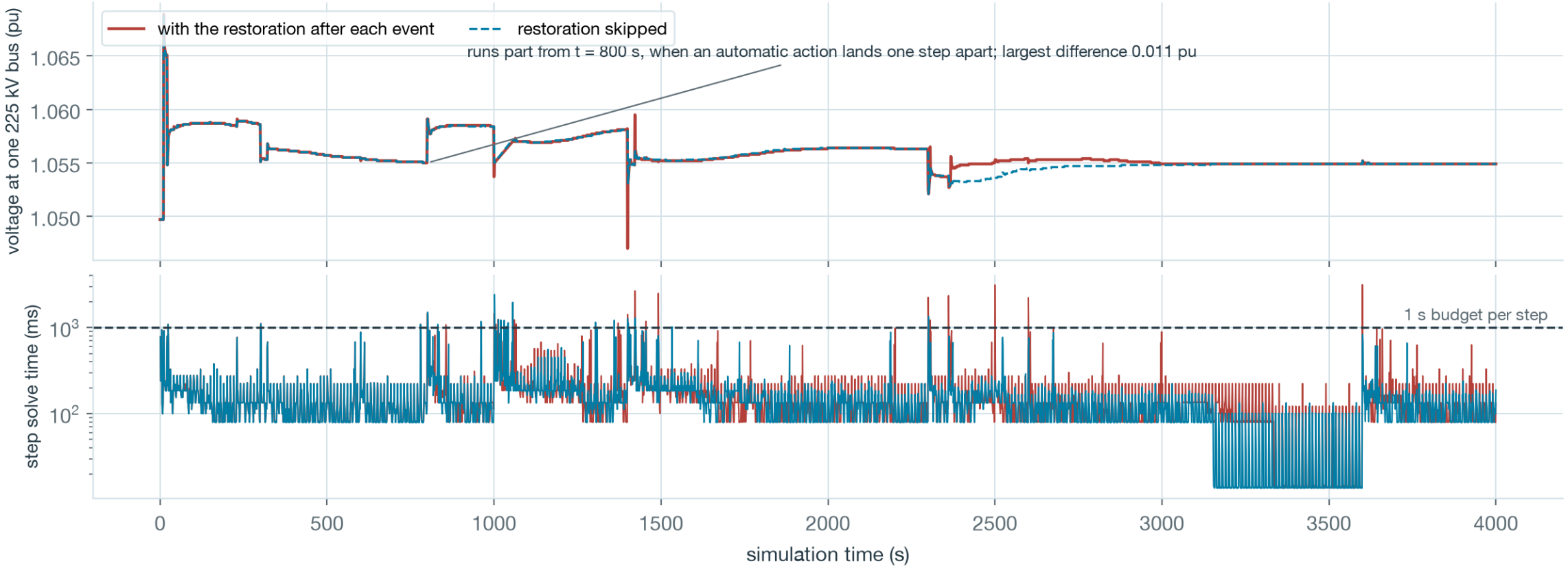
OP 20: The one late step, at t = 300 s, falls from 1751 ms to 965 ms

57 operating points	With the restoration	Restoration skipped
Late steps	309	239
Operating points fully in real time	0	**13**

What skipping the restoration changes in the result

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On a long scenarios with many events the two runs can differ by about 0.01 pu



4000 s scenario with eight events: an automatic action lands one step apart, 0.011 pu until the runs converge.

Solver status against the real-time goal

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Three improvements on one baseline, judged on late steps

Late steps per case	Reference	1. Fixed Jacobian pattern	2. Parallel evaluation, 8 threads, added to 1	3. Restoration skipped, added to 1
400 s benchmark, one scenario on the French transmission system	6	2	1	2
4000 s scenario, the same system with eight scheduled events	34	26	16	23
57 operating points, each run through the same six-event 400 s scenario	651	332	371	239
Operating points in real time on every step, of 57	0	0	0	**13**

- Only 13 of the 57 operating points are within the one-second real-time horizon .
- Improvement 1 is proposed to the open-source simulator project; 2 and 3 stay in the project's version: github.com/SPS-L/dynawo

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Conclusions

One problem, two routes:

Smaller model

Adaptive Model Selection

- Predicts where detail is needed from the initial state and the planned event alone.
- Removes 72 % of continuous and 66 % of discrete variables on the French test set.
- Zero late steps in all three illustrated cases but might lose accuracy where a critical substation is simplified.

Faster solver

Three solver improvements

- A fixed Jacobian pattern halves the late steps; proposed to the open-source project.
- Skipping the restoration after an event puts 13 of the operating points in real time.
- Parallel evaluation: 12 % more throughput, peak step improvement.

AMS reaches real time but can lose accuracy where a critical substation is wrongly simplified; the solver optimization route keeps the accuracy but is not yet within the one-second real-time horizon.

Three fronts, one milestone: the two routes together on the training platform

Model route

Adaptive Model Selection

- Add the control and communication links that trigger remote actions.
- Adapt device models, not only substation topology.
- More networks, disturbances and dynamic quantities.

Solver route

Numerical improvements

- Attack the numeric factorisation, the largest block of a late step.
- Restructure the network model, two thirds of the loop.
- Take the fixed pattern through open-source review.

Platform

Operator training

- Run both routes together on the same scenarios.
- Measure the complete platform latency during an operator exercise.

Achieve each simulated second in less than a second of real time without loss of accuracy

Project outputs and support

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Research outputs

Power Systems Computation Conference 2026

- **Presented**: Towards an Open-Source Real-Time Operator-Training Platform: Analysis of Computational Efficiency

Power System Dynamics Summit 2026, Best Paper Award

- **Presented**: Adaptive Reduced System Modeling for Real-time Dynamic Simulations

IEEE Transactions on Power Systems

- **Submitted**: Graph-Based Dynamic Activity Prediction for Real-Time Power System Applications

GitHub, Apache 2.0: github.com/SPS-L/DYNAGNN

- **Released**: DYNAGNN v1.2, activity prediction and model selection

Internal reports

1st technical report

- Initial Steps and Investigations

2nd technical report

- Profiling Dynawo Software

3rd technical report

- Adaptive Model Selection for Real-Time Operator Training Platforms

Support and collaboration

- **Funding:** TRAISIM, funded by CRESYM, project manager Geethu Joseph; TwinEU (101136119)
- **Partners:** SPS-Lab @ CUT and RTE (Thibaut Vermeulen)

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Backup slides

- Four practical questions: another network, where the software runs, a wrong prediction, the distance to real time
- Limitations
- Further results: variable counts, parallel evaluation, every operating point
- Method of the activity predictor: training data, encoders, activity classes, graph features, architecture, encoder choice

What it takes to use this on another network

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Data to train on, one training run per network, then a millisecond per scenario

- Operating points and a contingency list: 61 and 69,616 scenarios on the French system.
- Detailed offline simulations of those scenarios; the labels come from the simulated curves.
- One training run per network; about 1.4 ms per scenario at run time.
- Transfer to another network is not yet established.

Solver route: settings in the simulator. Model route: a step before the simulation

- Three solver settings in the project's version of the open-source simulator, off by default.
- The first is under review in the open-source project.
- The lighter model is written by a step before the simulation; the simulator is unchanged.
- That step is the open-source DYNAGNN package, installed with pip: github.com/SPS-L/DYNAGNN
- All timings: serial runs on one machine.

When the predictor is wrong

Two kinds of error, one dial, and what is not yet in place

- Too much detail kept: slower, still right. Too little: fast, a voltage wrong by 0.014 pu.
- The retention threshold is the dial: 73 % of variables kept at its most cautious setting, 28 % as shown.
- A predicted action always forces full detail.
- Not yet: an online check against the full model.

How far from real time, in seconds

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Per 400 s scenario on the French transmission system, 57 operating points

Per 400 s scenario, 57 operating points, pattern fixed	With the restoration	Restoration skipped
Late steps from the first event on	208	139
Time behind, total over the 57 scenarios	85 s	33 s
Time behind, average and median per scenario	1.5 s, 0.8 s	0.6 s, 0.1 s
Worst single pause after an event	2.2 s	1.9 s
Before any event: late steps, time, runs affected	101, 86 s, 35	100, 83 s, 35

- Behind: time above 1 s, summed over late steps; events are scheduled from t = 50 s.
- The last row is a solver defect: a failed model evaluation in the first seconds, then repeated step halving, up to 13 s on one run. Not an event cost; under investigation.

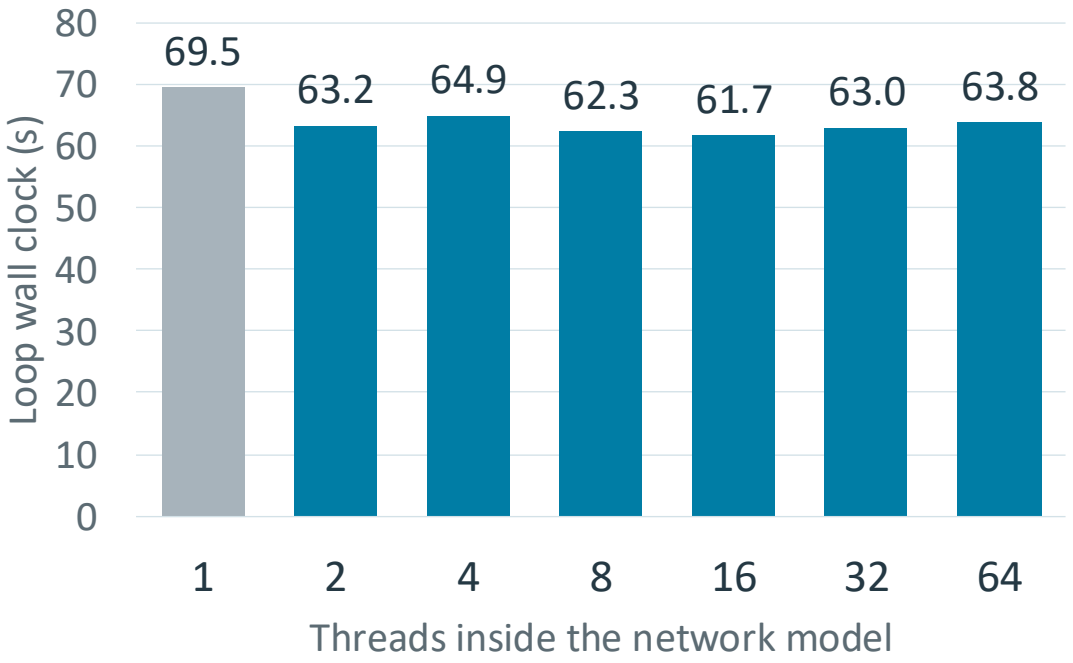
Limitations

- Performance on a new network or on unseen equipment identities is not yet established.
- The graph omits some remote control and communication relationships.
- Simplification currently changes switching topology while retaining dynamic device-model structures.
- Selected timing examples do not guarantee accurate real-time performance for every scenario.
- No full-detail scenario yet runs within the one-second real-time horizon on every simulated second; numeric factorisation, the largest block of a late step, has no solver item yet.

Parallel evaluation inside the network model

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Threads raise throughput but do not lower the peak step time



- Model evaluation is about two thirds of the solver loop, nearly all of it in one network object of 14,944 components.
- Three regions of it run in parallel; branch accumulation stays serial, so the curves are bit-identical.
- Wall clock falls 12 % at eight threads and the curve is flat up to 64 threads, but the peak step time rises by 150 ms.
- Effective parallel fraction near 12 %: the network holds 90 % of the Jacobian's nonzeros in one task.

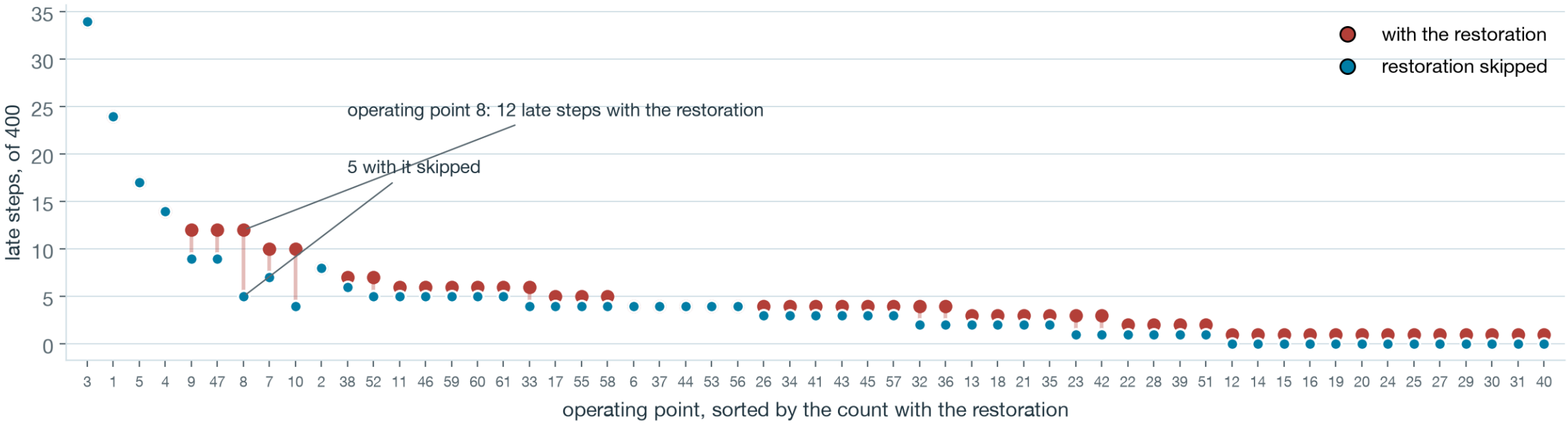
Kept in the project's version of the simulator, off by default: the goal is the peak step time, not throughput

400 s benchmark with the Jacobian pattern fixed, medians of five serial replicates per thread count. Parallel Jacobian assembly was built, measured at no effect and reverted.

Restoration skipped: all operating points

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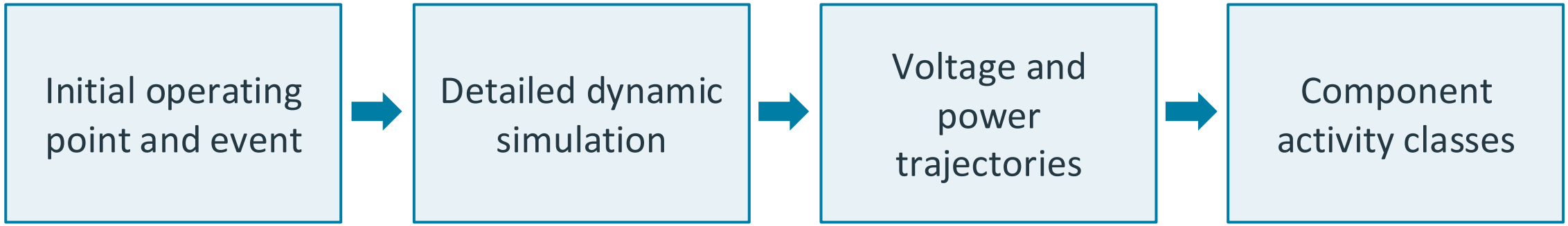
No operating point gets worse; the gain is largest where restorations are many



- 47 points improve, 10 are unchanged and none is worse; 13 reach real time. Operating point 8 goes from 12 late steps to 5.
- The heaviest points do not move: their late steps are not restoration steps. Four further points stop early in both configurations.

Late steps per operating point, 400 steps of one simulated second each, Jacobian pattern fixed in both configurations.

Complete simulations provide the activity targets (data) for learning



- Repeat across different loading patterns and network disturbances.
- Keep complete operating points outside training to test generalization.

At prediction time, only the initial state and planned event(s) are required.

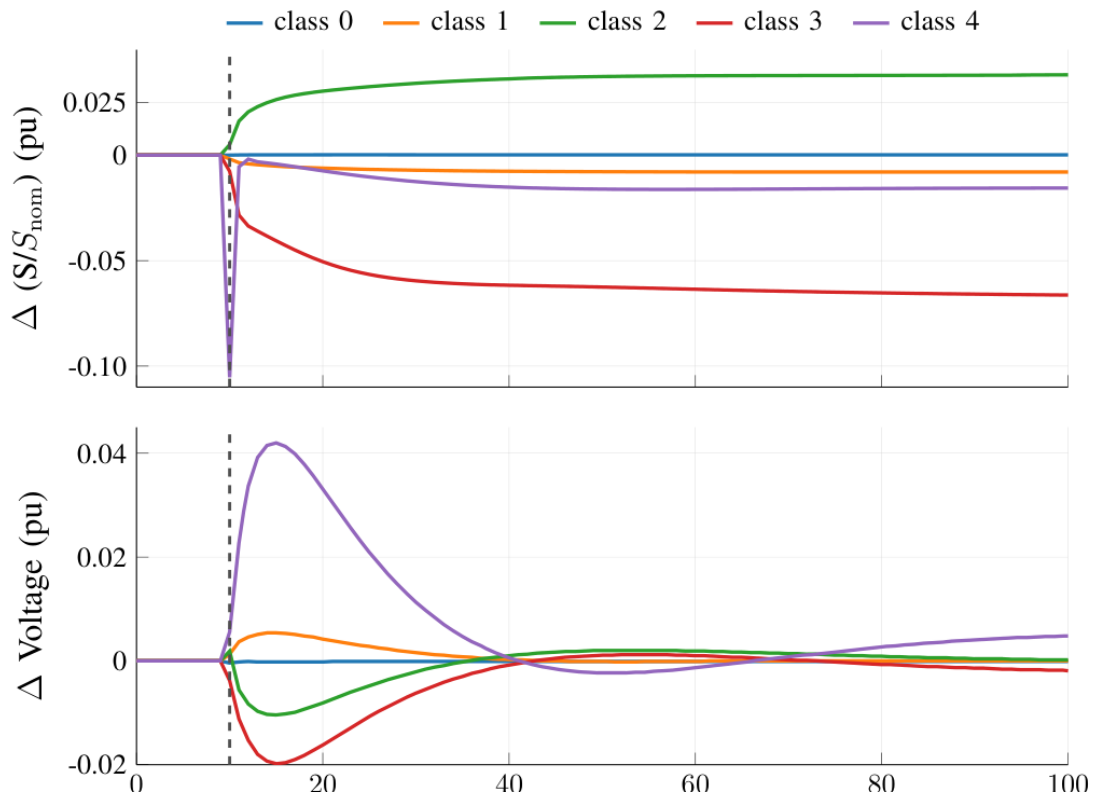
Predicting the response to a planned event

Dedicated encoders combine component, event and network context. GINE keeps graph computation low.

Encoder	Information	Purpose
Component	Type, identity and initial electrical state	Describe the target
Event	Affected element and event identity	Describe the disturbance
Graph (GINE)	Connections and edge attributes	Share network context
Target-event pair	Component/event embeddings and graph summary	Predict the response to this event

GINE: Graph Isomorphism Network with Edge features. An accuracy and cost comparison follows.

What does dynamic activity mean?



Maximum moving-window variance

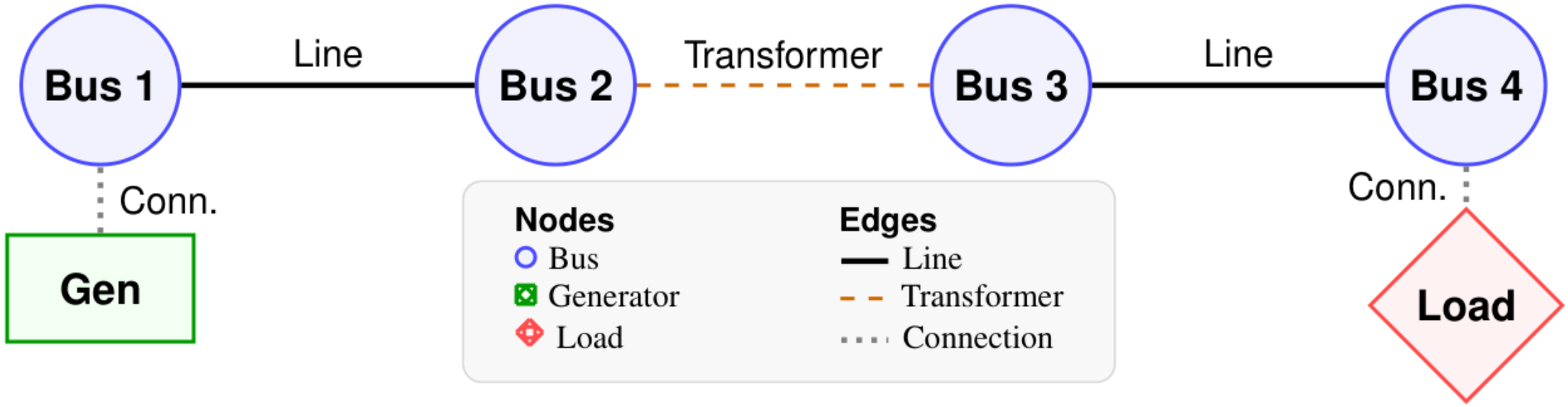
- Voltage in pu, or generator S/Snom.
- 5 s moving windows, with 1 s step.

Class	Meaning
0	Inactive within numerical tolerance
1–4	Increasing dynamic activity
5	Non-simplifiable action or disconnection

Inactive cut: $\tau_0 = 10v^2 = 10^{-7}$
v: numerical solver tolerance (10^{-4})

Higher cuts use training data only, balancing class usability with physical severity and scenario coherence.

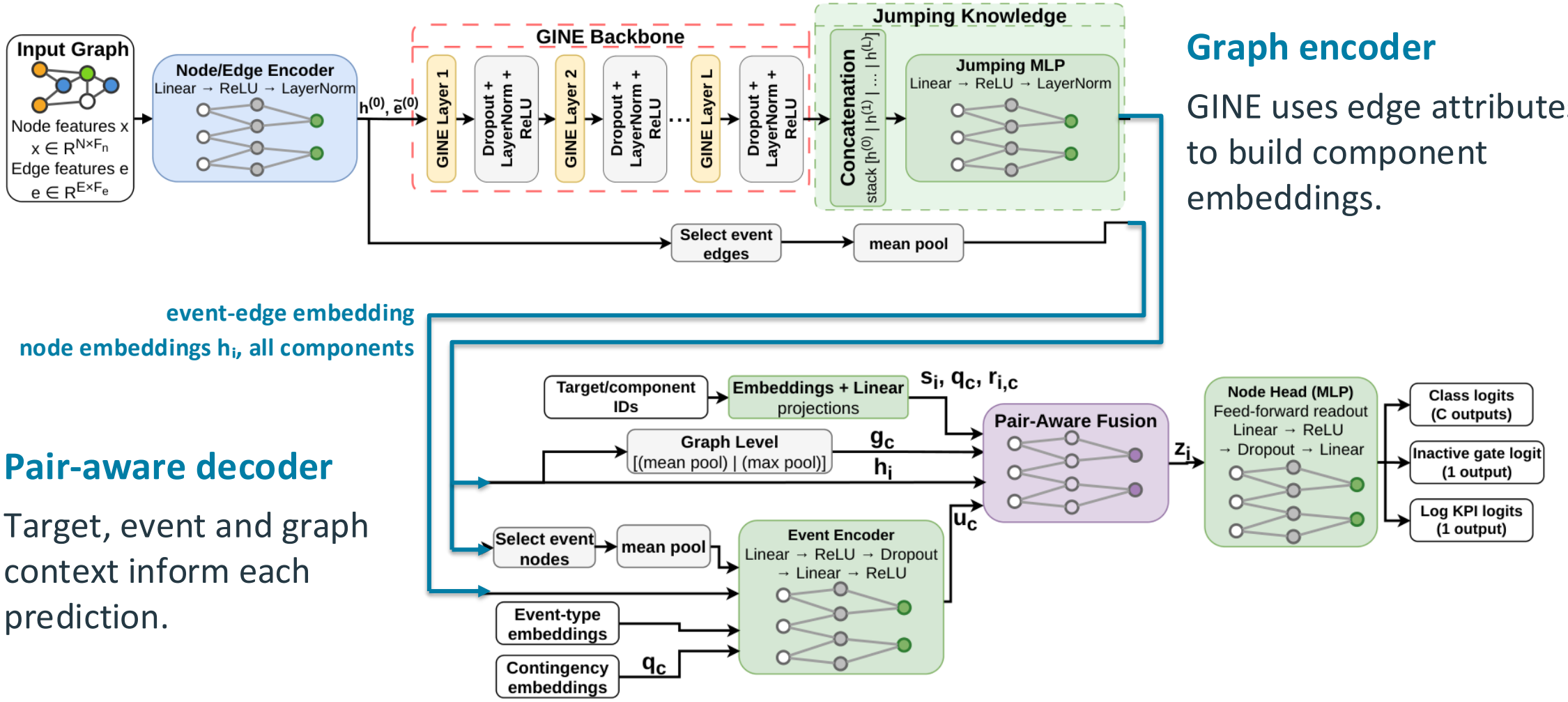
Representing the grid and disturbance



Graph information	Inputs to the predictor
Nodes	Bus, generator and load types, initial voltage, angle, P and Q
Edges	Connection type, event indicator and electrical parameters
Contingency coupling	Log-transformed transfer-impedance magnitude $\lvert Z_{ij} \rvert$

Transfer-impedance magnitude describes electrical coupling to the event location.

A graph model for component activity



Graph encoder

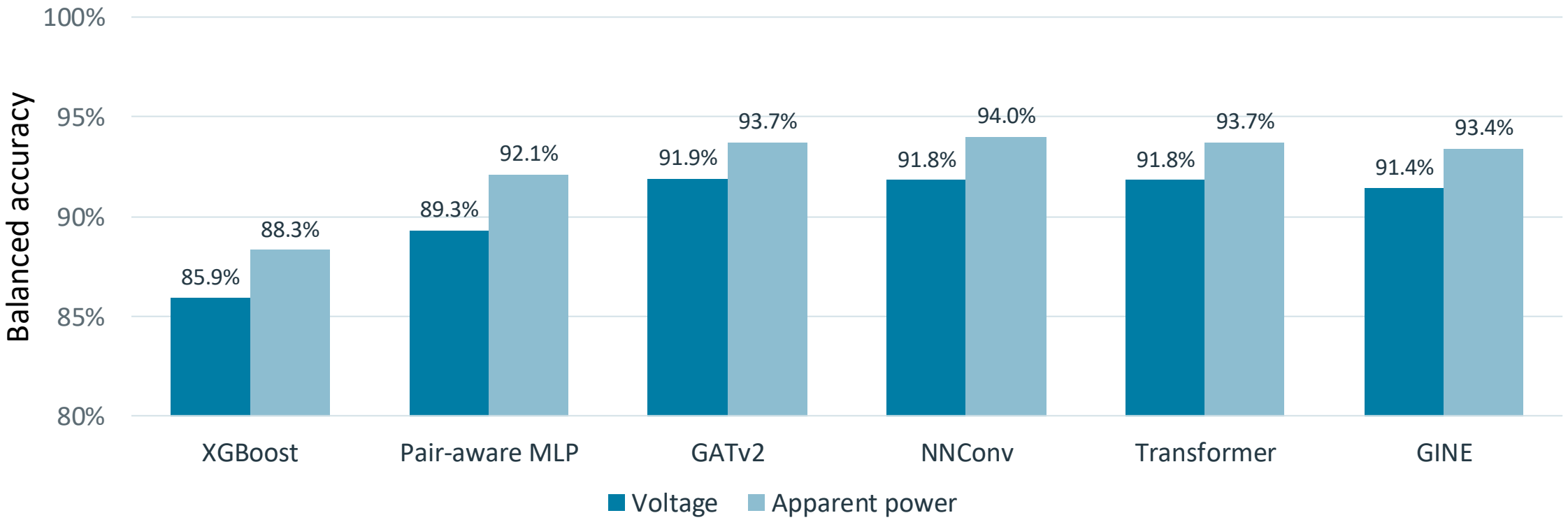
GINE uses edge attributes to build component embeddings.

Pair-aware decoder

Target, event and graph context inform each prediction.

Independent voltage and apparent-power models use the class decoder for the reported results.

Choosing an efficient graph encoder



GINE: 91.4% / 93.4% balanced accuracy at about 1.4 ms per task and scenario

Edge-aware encoders perform similarly. GINE has the fewest parameters and lowest median inference cost among them.